Tools for Data Visualization

BI304

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Why Use Data Visualization Tools?

- See patterns/outliers faster than tables
- Communicate insights to non-experts
- Support quicker, better decisions
- Align teams on a shared view of the data

Why Data Storytelling in Business

- Narrative turns insight into action
- Adds context, sequence, and relevance
- Improves recall and persuasion
- Bridges analysts and decision-makers

Data Visualization vs. Data Storytelling

- Visualization: charts/maps that show data
- Storytelling: narrative + visuals + message
- Visualization = evidence; Story = meaning
- Goal: help audiences understand and care

Tool Categories Overview

- Programming libraries (Python): Matplotlib, Seaborn, Plotly, Altair
- No-code/low-code: Datawrapper, RAWGraphs
- Geospatial: Leaflet, kepler.gl
- Dashboards (Python): Streamlit, Dash
- Storytelling: Flourish

Matplotlib — What It Is

- Foundational Python plotting library
- Static, publication-quality graphics
- Fine-grained control over every element
- Backbone for many higher-level tools

Matplotlib — Pros / Cons / Demos

- Pros: precise control, stable, huge ecosystem
- Cons: verbose, steeper learning curve, static
- Use cases: reports, journals, branded figures
- Demo idea: multi-series sales line with annotations

Seaborn — What It Is

- High-level statistical graphics on Matplotlib
- Clean defaults, quick attractive plots
- Tight pandas integration
- Specialized plots (heatmap, violin, pairplot)

Seaborn — Pros / Cons / Demos

- Pros: minimal code, great for EDA, readable styles
- Cons: limited deep customization; static
- Use cases: distributions, correlation, comparisons
- Demo idea: correlation heatmap for KPI set

Plotly — What It Is

- Interactive charts in Python (web-ready)
- Hover, zoom, pan by default
- Wide chart catalog incl. maps & 3D
- Works in notebooks and web apps

Plotly — Pros / Cons / Demos

- Pros: interactivity out-of-the-box; polished
- Cons: heavier for huge datasets; API depth
- Use cases: executive dashboards, live demos
- Demo idea: interactive revenue time-series with range slider

Altair — What It Is

- Declarative Python library (Vega-Lite grammar)
- Specify what to plot, not how
- Concise syntax; built-in interactivity
- Great for quick prototypes

Altair — Pros / Cons / Demos

- Pros: minimal code, consistent grammar, clean look
- Cons: less flexible for bespoke designs; data size limits
- Use cases: rapid exploratory charts, teaching gg-style thinking
- Demo idea: category sales over time + interactive legend filter

No-Code / Low-Code — Overview

- Build visuals via UI; no programming
- Fast turnaround; good defaults
- Ideal for non-technical stakeholders
- Limits on deep customization/automation

Datawrapper — What It Is

- Browser tool for charts, maps, tables
- Upload CSV/Sheet → choose template → publish
- Responsive, embeddable outputs
- Friendly for quick communications

Datawrapper — Pros / Cons / Demos

- Pros: very fast; clean designs; easy embedding
- Cons: manual data updates; template constraints
- Use cases: blog posts, press, internal memos
- Demo idea: grouped bar — quarterly revenue by region

RAWGraphs — What It Is

- Open-source for uncommon/custom charts
- Drag-and-drop data → map to visual channels
- Many advanced models (sankey, alluvial, dendrogram)
- Exports SVG/PNG for design polish

RAWGraphs — Pros / Cons / Demos

- Pros: unique visuals, designer-friendly, privacy in-browser
- Cons: static outputs; one-off workflows
- Use cases: standout infographics, research figures
- Demo idea: alluvial of traffic source → product → outcome

Geospatial — Why/When

- Location patterns: markets, stores, routes
- Layers: points, lines, polygons, heat/hex bins
- Reveal regional disparities and hotspots
- Essential for logistics, site selection

Leaflet — What It Is

- Lightweight JS library for web maps
- Add tiles, markers, popups with few lines
- Large plugin ecosystem (clustering, heatmaps)
- Works with Folium in Python

Leaflet — Use Cases / Demos

- Store locator, asset maps, incident maps
- Internal geo dashboards (ops, field teams)
- Demo idea: branch locations + KPI popup
- Tip: cluster markers for dense city data

kepler.gl — What It Is

- Web app for large-scale geo data (GPU)
- Drag-drop CSV/GeoJSON → instant map
- Layers: points, arcs, hex/grids, 3D extrusions
- Time playback for spatiotemporal patterns

kepler.gl — Use Cases / Demos

- Logistics, mobility, urban analytics
- Millions of points; quick filtering
- Demo idea: 3D hexbin of deliveries by hour
- Export sharable HTML or images

Dashboards (Python) — Overview

- Turn analyses into interactive web apps
- Inputs: sliders, dropdowns, date pickers
- Share with stakeholders in a browser
- Two paths: Streamlit (rapid), Dash (custom)

Streamlit — What It Is

- Script-to-app in minutes (st.* widgets)
- Auto-rerun on change; simple layout
- Easy chart display (Matplotlib/Plotly/Altair)
- Great for prototypes and internal tools

Streamlit — Pros / Cons / Demos

- Pros: fastest dev cycle; zero front-end needed
- Cons: limited UI control; simpler apps
- Use cases: model demos, data explorers
- Demo idea: campaign A/B explorer with filters

Dash — What It Is

- Plotly’s Python framework for dashboards
- Component/layout model; explicit callbacks
- Highly customizable; multi-page apps
- Suited for production deployments

Dash — Pros / Cons / Demos

- Pros: fine control, custom CSS/components
- Cons: more boilerplate; steeper learning curve
- Use cases: executive KPI portals, client-facing apps
- Demo idea: region selector updating multiple charts

Choosing Tools & Workflows

- Purpose: explore (code) vs present (story) vs operate (dashboard)
- Audience: technical vs non-technical
- Data: size (GPU?), geospatial?, real-time?
- Workflow: Python EDA → map/dash → storytelling export

Q&A